LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	17
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Flight control surface balancing
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Flight control surface balancing
5. Specific Objectives	6.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of control surface balancing procedure on training equipment; Safety: Ensure training equipment is secured; Supervision: Instructor monitors alignment and torque application
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of flight control surfaces?
(b) How does an unbalanced control surface affect aircraft handling?
(c) Why is balancing critical for high-speed flight performance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Causes of control surface imbalance • Effects of aerodynamic hinge moments • Types of balancing methods • Adjustment of counterweights • Verification of balance limits	L & Ds	PPT, IB	15 min
(b)	• Step-by-step balancing procedure • Use of balance tools and scales • Torque application standards • Inspection of hinge points • Documentation of adjustments	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting common balance issues • Impact on aircraft stability • Maintenance intervals for rebalancing • Safety checks post-balancing • Reporting discrepancies to maintenance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between hinge moment and control surface balance.
(b) Describe the procedure for verifying that a control surface is properly balanced.
(c) What are the consequences of neglecting control surface balancing during maintenance?

12. Home Assignment and Follow Up	Prepare a brief report on how control surface balancing principles apply to the next lesson on flow separation and stall behavior.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Flow Separation Stalling, behavior of aircraft before and after stall. Factors effecting stalling speed, high incidences stall, high speed stall, G stall, wing tip stall, Basic stalling speed.	Refer to the master syllabus reference list	LP 18

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min